MA'ADEN معادن

MA'ADEN ENGINEERING MANUAL

Pile Testing

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1. INTRODUCTION

1.1 Scope

This document specifies the minimum requirements for testing of piles for all Ma'aden projects. Testing comprises both weld testing on driven steel piles, and dynamic and static load testing of all pile foundations.

This document is intended to be used in conjunction with the applicable documents listed herein.

1.2 Purpose

The purpose of this specification is to set the minimum engineering standard for pile testing.

2. KEY TERMS AND DEFINITIONS

Cast In-Place Pile	A deep foundation unit made of cement grout or concrete and constructed in its final location, e.g. drilled shafts, bored piles, caissons, auger cast piles, pressure-injected footings, etc
Contractor	An entity, engaged or approved by the Owner, entrusted with a contract to perform work or cause work to be performed.
Deep Foundation	Referred to herein as a pile.
Driven Pile	A deep foundation unit made of pre-formed material with a predetermined shape and size and typically installed by impact hammering, vibrating, or pushing.
Employer	Saudi Arabian Mining Company (Ma'aden) and its authorized representatives and respective affiliates.
High Strain Dynamic Testing	This dynamic test method covers the procedure for applying an axial impact force with a pile driving hammer or a large drop weight that will cause a relatively high strain at the top of an individual vertical or inclined deep foundation unit, and for measuring the subsequent force and velocity response of that deep foundation unit. High-strain dynamic testing applies to any deep foundation unit, also referred to herein as a "pile".
ITP	Inspection and Test Plan.
Lateral Load Testing	This test method is used to measure the lateral deflection of a vertical or inclined deep foundation when subjected to lateral loading.
Low Strain Impact Integrity Testing	This test method covers the procedure for determining the integrity of individual vertical or inclined piles by measuring and analyzing the velocity (required) and force (optional) response of the pile induced by an (hand held hammer or other similar type) impact device usually applied axially and perpendicularly to the pile head surface.

MP	Magnetic Particle.
NDT	Non-destructive Testing.
PEM	Pulse Echo Method for low strain impact integrity testing.
RD	Radiography.
Static Axial Compression Load Testing	This test method is used to measure the axial deflection of a vertical or inclined deep foundation when loaded in static axial compression.
Static Axial Tensile Load Testing	This test method is used to measure the axial deflection of a vertical or inclined deep foundation when loaded in static axial tension.
TRM	Transient Response Method for low strain impact integrity testing.
UT	Ultrasonic Testing.
VE	Visual Examination.
VS	Visual Scanning.
Work	The entire work and Services and all things which the Contractor is or may be required to execute, supply and do under the Contract having regard to the Scope of Project, or to cause to be executed, supplied or done, for the purposes of constructing and Commissioning the Plant.

3. REFERENCED DOCUMENTS

This specification requires compliance with the provisions of the latest revision of the relevant standards and codes of the following authorities and technical organizations, as applicable, and specifications which shall be considered to form part of this specification, where applicable.

3.1 Statutory Regulations

All work performed and materials furnished shall comply with the applicable statutory regulations, codes and other requirements of the Kingdom of Saudi Arabia (including but not limited to directives of the High Commission for Industrial Security, Royal Commission regulations applicable to the Site, regulations of the Presidency of Meteorology and Environment, Saudi Building Code and other applicable codes and regulations). The Contractor shall ensure that he is fully aware and informed of all Government laws, local ordinances, regulations and all orders and decrees of bodies or tribunals having any jurisdiction or authority over the Project and/or Site, which in any manner affect the Project and those engaged or employed on the Project, or which in any way affect the conduct of the Work. The Contractor shall at all times observe and comply with all such Government and/or Ministry laws, bylaws, ordinances, codes, regulations, orders and decrees.

3.2 Industry Codes and Standards

All pile testing shall conform to the following.

3.2.1 Local Codes and Standards

SBC 303	Soil and Foundations Requirements

3.2.2 International Codes and Standards

3.2.2.1 American Society for Testing and Materials (ASTM)

ASTM A500	Standard Specification for Cold-Formed Welded and Seamless Carbon Steel Structural Tubing in Rounds and Shapes
ASTM D1143	Standard Test Methods for Deep Foundations under Static Axial Compression Load
ASTM D2487	Classification of Soils for Engineering Purposes (Unified Soil Classification System)
ASTM D3689	Standard Test Methods for Deep Foundations under Static Axial Tensile Load
ASTM D 3740	Standard Practice for Minimum Requirements for Agencies Engaged in the Testing and/or Inspection of Soil and Rock Used in Engineering Design and Construction
ASTM D3966	Standard Test Methods for Deep Foundations under Lateral Load
ASTM D4945-00	Standard Test Method for High Strain Dynamic Testing of Piles
ASTM D5882	Standard Test Methods for Low Strain Impact Integrity Testing of Deep Foundations

3.2.2.2 American Society of Mechanical Engineers (ASME)

ASME/BPVC	Boiler and Pressure Vessel Code

3.2.2.3 American Welding Society (AWS)

AWS B1.10M/B1.10	Guide for the Nondestructive Examination of Welds
AWS D1.1M/D1.1	Structural Welding Code – Steel

3.2.2.4 European Standards

European Standard NF EN 10219-1	Cold formed welded structural hollow sections of non-alloy and fine grain steels
European Standard NF EN 12699	Execution of special geotechnical work – Displacement piles
European Standard NF EN 10204	Metallic products – Types of inspection documents
European Standard NF EN 1011-1	Welding – Recommendations for welding of metallic materials – Part 1: General guidance for arc welding

European Standard NF EN 1011-2	Welding – Recommendations for welding of metallic materials – Part 2: Arc welding of ferritic steels
European Standard NF EN 287-1	Qualification test of welders , Fusion welding, Part 1: Steels
European Standard NF EN 1418	Welding personnel, Approval testing of welding operators for fusion welding and resistance weld setters for fully mechanized and automatic welding of metallic materials
European Standard NF EN 719	Welding coordination – Tasks and responsibilities
European Standard NF EN ISO 5817	Welding – Fusion-welded joints in steel, nickel, titanium and their alloys (beam welding excluded) – Quality levels for imperfections
European Standard NF EN ISO 4063	Welding and allied processes – Nomenclature of processes and reference numbers
European Standard NF EN ISO 15607	Specification and qualification of welding procedures for metallic metals – General rules

3.2.3 Alternative Codes

Alternative codes may be used for pile testing if approved by the Employer.

3.3 Ma'aden Standards

MD-101-SMEM-EG-CV-SPC-0001	Earthworks
MD-101-SMEM-EG-CV-SPC-0008	Topographical survey
MD-101-SMEM-EG-CV-SPC-0009	Geotechnical investigation
MD-101-SMEM-EG-ST-SPC-0001	Steelwork Fabrication
MD-101-SMEM-EG-GE-SPC-0002	Welding
MD-101-SMPM-PM-QA-PEI-0001	Contractor Quality Requirements

4. EXECUTION

4.1 Inspection and Testing

4.1.1 General

Inspection and Test Plans (ITPs) shall be submitted for approval, a minimum of 14 days prior to work commencing. The Employer reserves the right to mark comments, and to add witness and/or hold points to the ITPs.

Each driven pile sub-length shall be marked with a unique identification number, which shall be cross-referenced and used in combination with the pile number of which it forms a part of, to track all Quality Assurance information.

4.1.2 Weld Testing – Steel Piles

The Contractor shall be responsible for and bear the costs of all required inspection and testing in accordance with the contractual requirements. All testing of welds beyond visual scanning and examination shall be done by a registered testing authority independent of the Contractor. The Employer, at its discretion, may perform additional tests and inspections.

Results shall be presented in a manner that demonstrates compliance with the requirements of NF EN 10219-1 (7.2), ASTM A500, and any other applicable specifications and codes.

Weld testing and inspection criteria shall be in accordance with AWS B1.10M/B1.10 or NF EN 12699 unless noted otherwise. Quality levels for imperfections shall be in accordance with AWS D1.1M/D1.1 and NF EN ISO 5817, quality level D unless otherwise noted.

Non destructive testing of welds shall be performed as necessary to qualify welding procedures and to ensure the quality of finished welds. This may include:

- Surface examination, including visual scanning, visual examination, dye penetration and magnetic particle testing
- Ultrasonic examination and/or Radiographic examination.

Unless noted otherwise on the drawings, the minimum level of weld examination and testing shall be in accordance with the most stringent of Table 4-1 or AWS D1.1.

Table 4-1 : Minimum Weld Examination and Testing Requirements

Type of Weld	Location	Method of NDT*			
		VS	VE	UT/RD	MP
Longitudinal Butt Welds	Field	100%	100%	10%	10%
Transverse Butt Welds at Pile Section Joints	Field	100%	100%	10%	10%
All other Butt Welds	Field	100%	100%	10%	10%
All Fillet Welds	Field	100%	100%	10%	10%

Abbreviations for the methods of NDT are as follows:

VS Visual Scanning
VE Visual Examination
UT Ultrasonic Testing
RD Radiography
MP Magnetic Particle

Non-destructive testing shall be performed by an accredited testing authority which has been approved by the Employer. Results of non destructive testing shall be made available for examination by the Employer.

No weld bead may be coated before it has been examined and accepted.

To be able to correct the cause of any major defects as early as possible, the testing program shall involve full testing of the first major component or group of welds.

All faulty welds shall be removed, replaced, re-examined and retested. Any faulty welds and details of proposed weld repairs shall be brought to the attention of the Employer. If in the opinion of the Employer the proposed repair will induce excessive distortion or residual stresses in the fabricated members, then it shall be rejected and a replacement member fabricated at no cost to the Employer.

In any group of fabrications tested, if more than 10% of a weld group proves defective, the level of testing from Table 4-1 shall be increased to 100% for the welds already completed. If any of these welds prove defective, they shall be removed, replaced, and retested.

4.1.3 High Strain Dynamic Testing – Driven Piles

4.1.3.1 General

Dynamic testing shall be undertaken to monitor hammer and drive system performance, and to evaluate pile capacity and integrity. Dynamic testing shall conform to the requirements of ASTM D-4945-00 – Standard Test Method for High Strain Dynamic Testing of Piles.

Dynamic testing shall be undertaken on at least 10% of the piles to be driven unless agreed otherwise with the EMPLOYER prior to the commencement of piles installation. The EMPLOYER reserves the right to ask for different number of piles to be tested at any stage during piles installation. In such a case a minimum 48 hours notice will be given.

Dynamic testing shall also be undertaken on the first two piles to be driven by each new hammer to the project, and the first two piles after each modification to each hammer (including drive system modifications), as required by Clause 3.5.3 of ASTM D-4945-00.

4.1.3.2 Equipment and Personnel

The dynamic testing shall be performed using an appropriate Pile Driving Analyzer Model such as PAK or PAL, or equivalent as approved by the Employer.

An experienced specialist consultant, who has achieved an Advanced Level or better on the Foundation QA Examination for Providers of PDA Testing Services (or equivalent qualification as approved by the Employer) shall be in charge of the Pile Driving Analyzer (PDA) operation, result interpretation, CAPWAP analysis, and reporting.

At least two strain transducers and two accelerometers shall be used, and any holes required in the permanent works part of the steel pile for fixing of these gauges shall be completely filled with weld, ground flush, and the protective treatment repaired on completion of testing.

The proposed testing company's name, the proposed testing equipment, and curricula vitae and certification of qualification of staff proposed to undertake the testing shall be submitted to the Employer at least four weeks prior to commencement of testing.

4.1.3.3 Testing and Reporting

The Pile Driving Analyzer shall be used to undertake stress wave measurement and Case Method Dynamic Pile Monitoring for the full driven length. A field report summarizing the dynamic testing results shall be submitted to the Employer on the same day that the pile is driven. As a minimum, this shall include:

- Pile details, including pile number and location
- A record of blow count for the entire driven penetration
- Maximum, minimum and typical driving stresses
- Maximum, minimum and typical transferred energy
- Elastic compression at the end of drive
- Set (penetration per blow) at the end of drive
- Estimated driving resistance at the end of drive, and
- Discussion on variation from previous trends in the dynamic test data.

Where restrike tests are nominated, they shall be made no sooner than 48 hours, or the otherwise nominated restrike interval, after initial driving. For scheduling purposes the restrike interval shall be assumed to be over consecutive working days. The restrike test shall be conducted with a hammer which has been warmed up by driving another pile and shall comprise of PDA monitoring for the first 5 or 10 blows delivered to the pile. A representative blow shall then be specified from these blows by the PDA operator for CAPWAP analysis if required. Testing and field reporting for restrike tests shall be as above.

A representative blow from either restrike testing or end of drive testing shall be selected for CAPWAP analysis by an experienced PDA operator. The Case Pile Wave Analysis Program (CAPWAP) shall be used to determine and report upon:

- Static tension and compression load capacity
- Soil resistance distribution
- Pile tensile and compressive stress as a function of set, and
- Pile displacement as a function of load.

A dynamic testing report shall be prepared for each dynamically tested pile and submitted to the Employer no later than five days after each dynamically tested pile is driven, which includes, as a minimum:

- Field report(s) as described above
- Discussion on hammer and driving system performance (efficiency, comparison to expected performance for the make and model of hammer used, comparison to other similar hammers, suggestions for improvement, etc)
- Discussion on driving stress levels and pile integrity

- CAPWAP analysis including the results as described above, and identification of the test data used for the analyses, and
- Summary table of all dynamic test results undertaken on the project to date.

4.1.4 Low Strain Impact Integrity Testing – Cast-in-Place, Driven Concrete and Concrete Filled Pipe Piles

4.1.4.1 General

Low Strain Impact Integrity Testing shall be undertaken to evaluate pile integrity for cast-in-place concrete pile foundations, driven concrete and concrete filled pipe piles . Dynamic testing shall conform to the requirements of ASTM D-5882-07 – Standard Test Method for Low Strain Impact Integrity Testing of Deep Foundations.

Dynamic testing shall be undertaken on 100% of all concrete piles or 50% with conditions as agreed between the Contractor and the Employer prior to the commencement of pile installation. The Employer may direct additional piles to be dynamically tested, or alternatively may direct the dynamic testing requirement for certain piles provisionally nominated so to be deleted. The Employer will provide such direction at least 48 hours prior to the scheduled driving of the pile in question.

4.1.4.2 Equipment and Personnel

The Low Strain Impact Integrity Testing shall be performed in accordance with ASTM D-5882-07, by a competent personnel meeting D3740 (Practice for Minimum Requirements for Agencies Engaged in the Testing and/or Inspection of Soil and Rock as Used in Engineering Design and construction), or equivalent as approved by the Employer.

The proposed testing company's name, the proposed testing equipment, and curricula vitae and certification of qualification of staff proposed to undertake the testing shall be submitted to the Employer at least four weeks prior to commencement of testing.

4.1.4.3 Testing and Reporting

Cast-in-place, driven concrete and concrete filled piles shall not be tested sooner than seven days after casting, or after concrete strength reaches 75% of its design strength, whichever is earlier.

Testing to be performed in accordance with ASTM D-5882-07.

A field report summarizing the testing results shall be submitted to the Employer on the same day that the pile is driven. As a minimum, this shall include:

- Pile identification, nominal and actual pile head diameter and length
- Date and method of concrete placement or pile driving
- As-built geometry that is, total concrete volume, nominal or actual diameter versus length, permanent or temporary casing, steel reinforcement, etc.
- Soil stratigraphy
- Any specific observation related to each pile tested that affects the pile construction, excavation, integrity, etc

- Location of transducers at pile head and corresponding measurements, and
- Date of pile test.

A Low Strain Impact Integrity testing report shall be prepared for each tested pile and submitted to the Employer no later than five days after each tested pile is installed, which includes, as a minimum:

- Field report(s) as described above
- General:
  - Project identification/location
  - Nearest geotechnical bore log information and location.
- Pile installation equipment or driving method
- Test piles:
  - Test pile identification
  - Test pile type and dimensions
  - Test pile date made, cast or installed or concrete cylinder strength or both
  - Description of internal and external reinforcement
  - Description and location of splices where applicable
  - Pile observations including spalled areas, cracks and head surface condition.
- Pile installation:
  - Date of installation
  - For drilled shafts, size of auger and volume of concrete or grout placed in pile, grout pressure used, and description of special installation procedures used such as pile casing installation or extraction
  - For driven piles (when driving is the suspected cause of damage), hammer cushion and pile cushion information, driving records including blow count and hammer stroke or operating level for final unit penetration
  - Cause and duration of interruptions in pile installation if applicable and related to the investigation
  - Notation of any unusual occurrence during installation or excavation, or both, that may relate to the integrity investigation.
- Integrity testing:
  - Description of the apparatus for obtaining impact measurements and apparatus for recording reducing and displaying data, and of test procedure including description and location of the motion sensor attachment

- Date tested
- Test pile identification
- Graphical presentation of motion and force (optional) measurements in the time domain which shall include data enhancements used in the evaluation; frequency domain presentation is optional for PEM evaluation but required for TRM evaluation
- The wave speed of the test pile and how determined or estimated
- Length of pile as driven or cast, embedded, or below apparatus for obtaining dynamic measurements
- Methods used to evaluate data
- Comments on the integrity of the pile, including when appropriate how the soil stratigraphy or installation methods, or both, have influenced the constructed shape as determined by the measurements.

4.1.5 Static Testing

4.1.5.1 General

Static testing shall be undertaken to evaluate axial and lateral pile capacity and shall conform to the requirements of:

ASTM D1143	Standard Test Methods for Deep Foundations under Static Axial Compression Load
ASTM D3689	Standard Test Methods for Deep Foundations under Static Axial Tensile Load

Lateral pile capacity shall conform to the requirements of:

ASTM D3966	Standard Test Methods for Deep Foundations under Lateral Load

Static testing shall be undertaken on 3% of all piles or as agreed between the Contractor and the Employer prior to the commencement of pile installation. The Employer may direct additional piles to be statically tested, or alternatively may direct the static testing requirement for certain piles provisionally nominated to be deleted. The Employer will provide such direction at least 48 hours prior to the scheduled driving of the pile in question.

4.1.5.2 Equipment and Personnel

An experienced specialist consultant shall be in charge of the static load test operation, result interpretation, analysis and reporting.

The proposed testing company's name, the proposed testing equipment, and curricula vitae and certification of qualification of staff proposed to undertake the testing shall be submitted to the Employer at least four weeks prior to commencement of testing.

4.1.5.3 Testing and Reporting

The Contractor shall prepare and submit a Method Statement for Static Load Testing including drawings of the proposed static test pile arrangement to the Employer a minimum of four weeks prior to commencement of testing. Testing equipment calibration curves and tables shall be submitted to the Employer as part of the Method Statement.

Each static load test record shall be submitted to the Employer within 24 hours of testing, and as a minimum shall include:

- Pile details, including pile number
- Details of the pile testing including testing type, configuration, loading mechanism, loading and unloading cycles etc
- Movements in reaction piles or reaction frame and corresponding loading
- Movement in test pile and corresponding loading
- Pile displacement as a function of load
- Pile displacement as a function of time, and
- Interpretation and discussion of static test pile results
- Recommendations.

5. QUALITY CONTROL

The Contractor shall refer to the following.

MD-101-SMPM-PM-QA-PEI-0001 Contractor Quality Requirements